

Acu.ation: Real Time Acupoint Stimulation To Mediate The Urge To Smoke.

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Figure 1: Left: Version 01 of prototype in use. Right: Version 02 of prototype.

ABSTRACT

Acu.ation is an intervention that aims to mediate the urge to smoke by pairing a wearable device that delivers transcutaneous electric acupoint stimulation (TEAS) with a mobile application that provides Mindfulness Based Relapse Prevention (MBRP) (Figure 1). Given the pervasiveness of cigarette smoking, high rates of relapse, and limitations of existing treatments, it is critical to explore new methods of relapse prevention. Our solution draws from acupuncture's global use in treating addiction and recent evidence that non-invasive TEAS can reduce the urge to smoke after exposure to a drug-related cue. Specifically, we have designed a device formed to four acupoints (LI4, PC8, PC6, and TH5) located on the wrist and hand, and programmed to deliver 5-15mA stimulation. Through a paired mobile application, individuals are simultaneously guided through an MBRP intervention to help them take control of their response to the trigger. Together, this system provides real-time relief from the urge to smoke so individuals can better engage with cognitive behavioral relapse prevention strategies in high risk moments.

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CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Interaction devices;

KEYWORDS

Addiction, Cue Reactivity, Transcutaneous Electric Acupoint Stimulation, Mindfulness Based Relapse Prevention, Cognitive Behavioral Therapy, Substance Abuse, Relapse Prevention, Cigarette Smoking

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1 INTRODUCTION

The use of tobacco products is pervasive across the United States. In 2018, the CDC National Health Interview Survey ($n = 25,417$) found that 19.7% of US adults use tobacco products with the majority using cigarettes (13.7%) [21]. Tobacco use rates are 5-15% higher among historically marginalized populations, including individuals who are non-Hispanic American Indian/Alaska native, lesbian, gay or bisexual, uninsured or insured by Medicaid, have a disability, or have experienced serious psychological distress [21]. Tobacco use, especially cigarette smoking, is also notoriously difficult to quit, as highlighted by the fact that despite 55.1% of smokers attempting to stop smoking at least once in the past year, only 7.5% successfully quit [21]. With cigarette smoking causing approximately one in

five deaths in the United States each year, this is a cause of great concern [32].

One major contributor to relapse among smokers is an individual's reaction to environmental cues. These can be proximal cues, stimuli directly related to the drug or drug paraphernalia, or distal cues, stimuli related to contexts in which the drug has been used [5]. Studies have shown that when smokers are exposed to such environmental cues, they report higher urge to smoke, and experience measurable changes in physiology and brain activation patterns. This reaction is believed to be a learned association between the cue and the drug [29]. Studies that monitor individuals longitudinally (in near real-time) after smoking cessation have also found that environmental cues are a significant factor in relapse [26, 27]. Furthermore, the strength with which an individual responds to an environmental cue has been shown to correlate with how likely they are to successfully quit smoking, thus supporting the idea that a person's cue-reactivity may be a mediating factor between cue exposure and relapse [29].

Existing relapse prevention methods for smoking consist of pharmacological and cognitive-behavioral interventions, each of which have significant limitations. In general, pharmacological interventions have not shown success in reducing cue-reactivity [29]. Additionally, a meta-analysis examining the efficacy of nicotine replacement therapy, the most widely used pharmacological intervention for smoking, found that it did not reduce relapse rates when compared to a placebo control [13]. On the other hand, some studies have found that using cognitive and behavioral strategies in moments of temptation decreases the likelihood of relapse [18, 28]. Mindfulness-based relapse prevention (MBRP) strategies appear particularly well-suited to decreasing relapse when an individual encounters an environmental cue, as they train individuals to be aware of and suppress their automatic reaction to drug-related triggers [14]. A meta-analysis by Li et al showed that mindfulness-based treatments significantly reduce rates of relapse among smokers [34]. Meanwhile, Bowen & Marlatt exposed participants who had received mindfulness instruction to a smoking-related cue and found that it did not reduce cue-reactivity in the moment [25]. These participants did smoke significantly less in the subsequent week, indicating that these cognitive strategies impact how individuals handle their reaction to a cue, not the strength of the reaction itself [25]. This demonstrates that the use of MBRP strategies is therefore limited to when individuals feel they can tolerate the strength of their reaction to a cue, which may be particularly difficult in the early stages of smoking cessation.

The need for relapse prevention strategies that mitigate the strength of cue reactivity motivated new work to be done using transcutaneous electric acupoint stimulation (TEAS). TEAS is a form of acupuncture in which individuals receive electrical stimulation on acupoints using non-invasive skin electrodes [3]. The use of TEAS has shown promising opportunities to reduce cue-reactivity in real-time, so that individuals may engage with helpful cognitive strategies. Multiple studies illustrate that actuation of TEAS on specific acupoints reduces self-reported urges to use as well as decreases physiological changes when an individual is exposed to a drug-related cue [3, 10, 16]. This suggests that TEAS can provide immediate relief in the moment of cue-reactivity or immediately following exposure to a cue.

Positioned at the intersection of these novel methods for relapse prevention, Acu.ation is the first intervention which we know of that combines MBRP techniques with TEAS acupuncture in a wearable device and mobile application that provides in-time assistance for individuals struggling with cue reactivity while trying to quit smoking. We hypothesize that TEAS delivered in the moment of cue-reactivity will decrease an individual's self-reported urge to smoke, as well as decrease their immediate physiological response. In addition, when TEAS is paired with MBRP instruction, individuals will be more likely to practice helpful cognitive behavioral techniques and will be less likely to relapse.

2 BACKGROUND

2.1 Transcutaneous Electric Acupoint Stimulation (TEAS)

In rat models, stimulation at specific acupoints significantly reduced ethanol and morphine self-administration when compared to stimulation at a non-acupuncture point or stimulation at an unrelated acupuncture point [31]. In a randomized placebo-controlled study done with human subjects, Yang et al also found that acupuncture at specific acupoints can increase people's ability to abstain from alcohol use [4]. While acupuncture's mechanism of action is not entirely understood, there is some evidence that acupuncture can suppress drug-seeking by modulating the mesolimbic dopamine (DA) system [30, 31]. The DA system plays an important role in the desire to seek safety and positive experiences, but excessive arousal of this system can cause feelings of urgency and craving [2].

TEAS, a non-invasive form of acupuncture, capitalizes on the benefits of acupuncture while expanding its potential applications. TEAS can be administered by the patient themselves, and does not need to be done in a lab-setting or in the presence of an acupuncture professional [3]. Research with rat models has shown TEAS to be at least as or more effective than electric acupuncture, which involves both electric stimulation and traditional acupuncture needles [17].

Multiple randomized placebo-controlled studies have shown that 2 Hz or 2/100 Hz of TEAS on the LI4, PC8, PC6, and TH5 acupoints reduces self-reported urges to use as well as decreases physiological changes when an individual is exposed to a drug-related cue [3, 10, 16]. LI4, also known as Hegu, is at the midpoint of the second metacarpal bone on the radial side. PC8, Laogong, is at the midpoint of the second and third metacarpals on the palmar side [7]. PC6, Neiguan, is 2 inches below the palm between the palmaris longus tendon and the flexor carpi radialis tendon. TH5, Waiguan, is 2 inches below the skin crease on the back of the wrist, between the ulna and radius bones (Figure 2) [7].

Interestingly, a placebo-controlled study that delivered 2/100 Hz TEAS on the LI4, PC8, PC6, and TH5 acupoints to cocaine-dependent and cannabis-dependent individuals separately from a related drug cue (at least 1 hour later) did not find an association between TEAS and decreased cue-reactivity [7]. This suggests that TEAS can provide immediate relief in the moment of cue-reactivity or immediately following exposure to a cue, but will not have an effect after that.

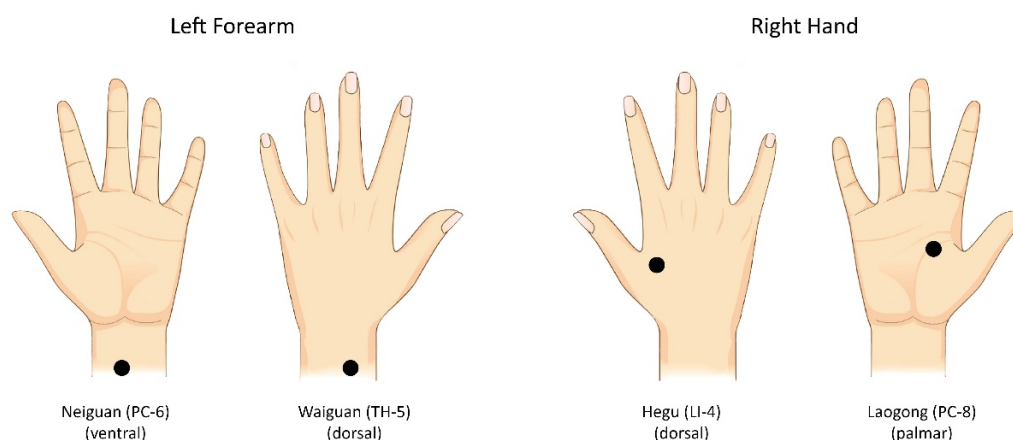


Figure 2: Diagram of acupoint locations for TEAS, from Penetar et al [3].

2.2 Mindfulness Based Relapse Prevention (MBRP) methods

MBRP was originally designed to be an eight-week out-patient treatment course. Each week involves a 2 hour meeting in which participants do formal meditation practices, group activities, and informal mindfulness exercises. In addition, participants are asked to do formal meditation practices and informal mindfulness exercises as homework between the meetings. One core practice that is introduced is called SOBER, which is meant to be used when a participant is triggered by an environmental cue. SOBER is an acronym that stands for Stop, Observe, Breathe, Expand and Respond and aims to help participants manage their reaction to the cue and take control of how they respond. Participants are first prompted to Stop, pausing in the moment they notice they are reacting. They are then instructed to Observe their thoughts, emotions and physical sensations. Next, with Breathe, they are asked to settle inwards on the sensation of breathing. Then they are prompted to Expand, becoming aware of their entire body, experience and context. Finally, they are encouraged to Respond, being aware of what they really need in the situation and doing what is best for themselves. In this way, participants are guided through a process of responding, rather than reacting, to the cue that triggers them [33]. Our proposed intervention will guide participants through this SOBER method via a mobile application when they experience the urge to smoke.

2.3 Real-time digital interventions for smoking cessation

The impact of in-time interventions has been greatly explored in recent years due to the growing capabilities of technology to readily capture, process and analyze personal data. Recent studies such as the ongoing SNAPSHOT study demonstrate the potential of smartphone and wearable technologies to deliver in-time interventions [1]. Similarly, Costa et al. have shown promising results for a wearable smartwatch to regulate emotions in the moment using vibrations that mimic one's heart rate [12]. In the context of smoking cessation, multiple studies are exploring ways to deliver CBT or

mindfulness strategies in the moment when a person experiences a trigger or urge to smoke [9, 11, 20, 24]. However, to our knowledge, no interventions have combined cognitive behavioral strategies with haptic methods of decreasing arousal in the moment of cue-reactivity. We believe that doing so will increase participants' ability to engage with the cognitive behavioral strategies and increase the efficacy of the intervention overall.

3 PROTOTYPE

Promising results from recent studies on the use of TEAS, cognitive behavioral interventions, and in-time digital interventions to reduce cue-induced urges and rates of relapse motivated us to design a digital system that pairs an app-based cognitive behavioral intervention with a wearable device that provides in the moment relief through the delivery of TEAS.

3.1 System Design

The system design consists of two components. First, the cognitive behavioral intervention, which is delivered through a mobile application that walks the user through the SOBER practice for relapse prevention. Second, the TEAS, which is delivered through a wearable device (Figure 3). Both of these treatments only begin once the individual is wearing the device and voluntarily activates the treatment through the mobile application.

Acu.ation is specifically designed to be used when an individual experiences the urge to smoke, triggered by environmental cues such as geographic location, social settings, and sensorial response. When they experience the urge to smoke, it is suggested that the individual finds a quiet place to sit, such as a park bench or bathroom, and opens the mobile application on their phone. They then can initiate the treatment through the mobile application, which starts the delivery of TEAS through the wearable device. Simultaneously, the mobile application verbally and visually guides them through the SOBER relapse prevention technique discussed above in the Mindfulness Based Relapse Prevention section (Figure 4).

The interface of the mobile application aims to be simple and intuitive. This allows users to quickly and easily start, stop or adjust

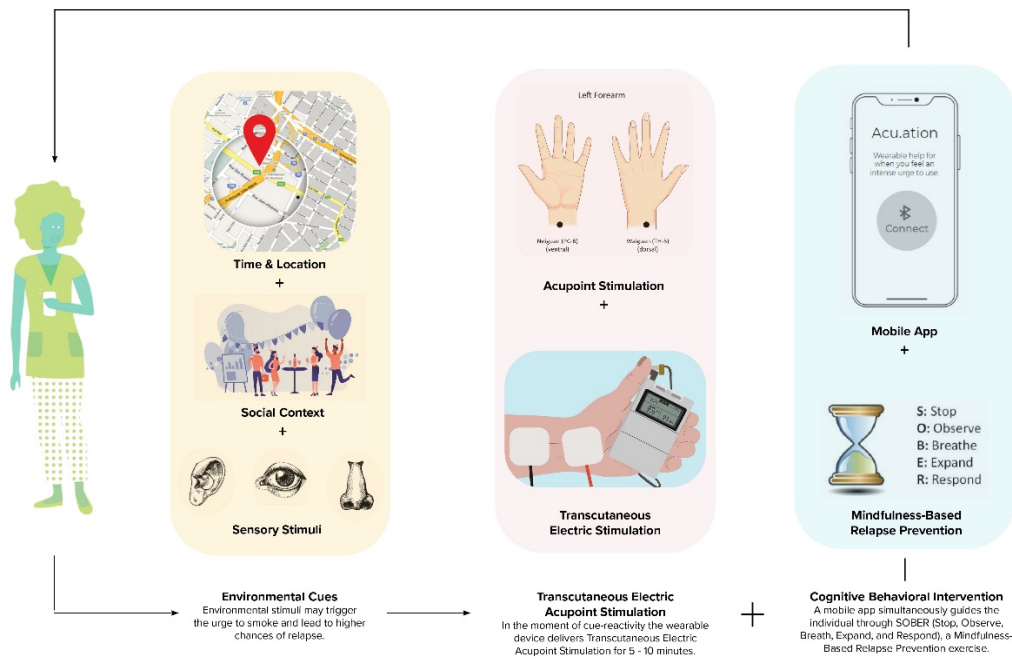


Figure 3: System design diagram.

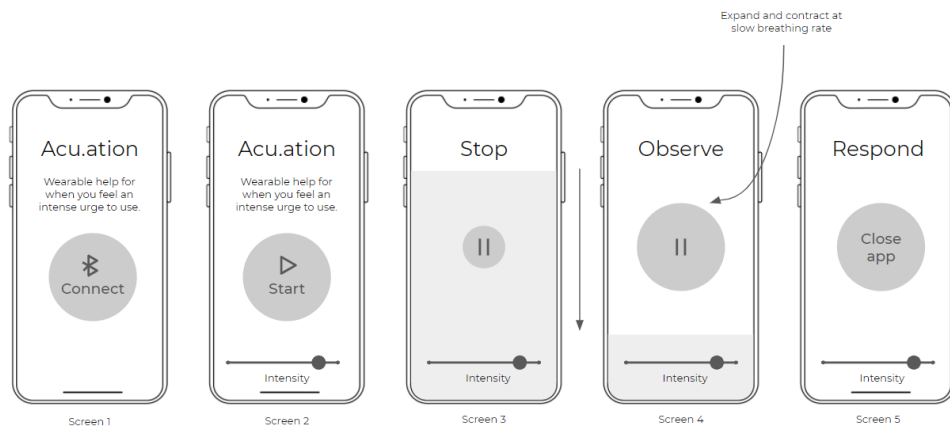


Figure 4: Mobile application user interface.

the treatment. It also ensures that once they begin the treatment, they can focus on the introspective SOBER technique without getting distracted by the application itself. Once the individual has completed the full sequence of the SOBER technique, which will range between five to ten minutes, the device will power off. Individuals may stop or increase/decrease the intensity of the TEAS at any point during the treatment.

3.2 Prototype Design

The wearable prototype delivering the TEAS is composed of four layers (Figure 5). First, two typical 1" circular TENS unit electrodes

are placed on the skin at each acupoint location. The second layer consists of a flexible smooth thermoplastic elastomer (TPE) which connects to the electrodes via two pieces of velcro as well as provides a platform for the electronics to sit on. The next layer consists of the hardware itself. Lastly, a 3D printed multi material structure composed of rigid and flexible thermoplastics envelops the previous three layers providing both a housing for the electronics while also making the wearable more aesthetically discrete.

The hardware of the piece relies primarily on a 1:3:10 adapter transformer, an NPN transistor, an Arduino Nano 33 IoT, and is supplied by a 7.4V lipo battery (Figure 6). The schematic for the circuit

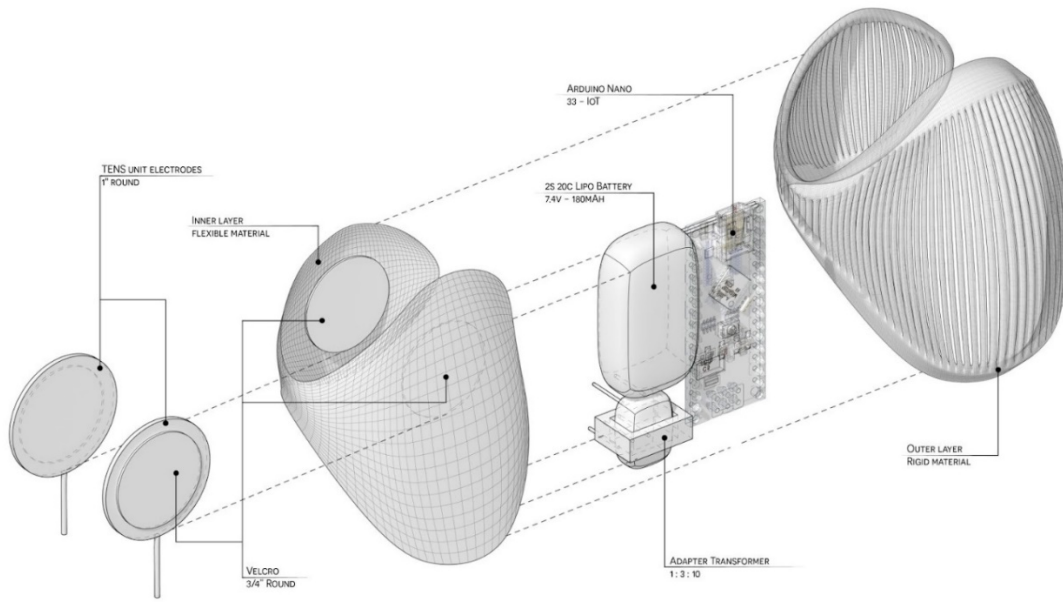


Figure 5: Exploded diagram illustrating each layer of the wearable prototype.

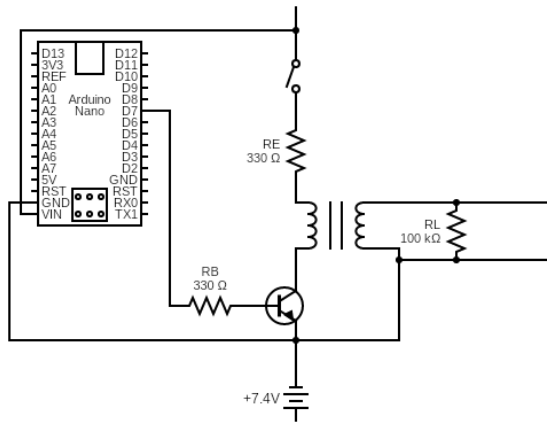


Figure 6: Schematic diagram of Acu.ation circuit.

was inspired by previous work done developing transcutaneous electrical nerve stimulation based on arduino and remote control using a smartphone [22]. This system was simplified and altered to meet the required settings of the device. These settings are that of the pulse amplitude, pulse width, and pulse rate which were sampled based on previous studies that found successful results in suppressing cue-induced tobacco urges and heroin cravings [3, 10].

Based on findings from these preliminary studies we proceeded in using the following settings for the TEAS. Pulse Amplitude of 10mA, which represents the intensity level of the electrical stimulation, a Pulse Width of 0.6ms, which indicates the length of each

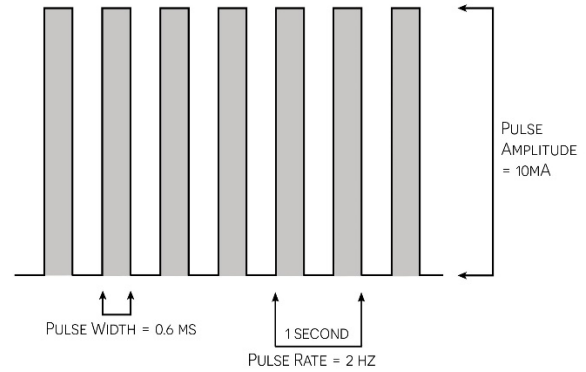


Figure 7: Diagram of Acu.ation device pulse settings.

pulse, and Pulse Rate of 2 Hz, which is also known as the number of times a pulse occurs in one second (Figure 7).

4 DISCUSSION

Existing relapse prevention treatments do not sufficiently reduce the urge to smoke when an individual is triggered by an environmental cue [13, 25, 29]. Even with cognitive behavioral therapies showing efficacy in reducing rates of relapse, there remains a great need for a real-time intervention which can help reduce cue-reactivity so individuals experiencing strong urges to use can engage with helpful cognitive behavioral techniques [15, 18, 25, 34]. This motivated the design of a system that pairs proven cognitive behavior techniques with novel methods of acupoint stimulation to provide relief from cue-reactivity and encourage the use of effective cognitive behavioral strategies. We envision this treatment not as a replacement for

current relapse prevention interventions, but rather as an addition that can easily be integrated into a person's treatment plan.

While the initial system is designed as a self-actuated interface, in which individuals would need to actively engage with the mobile app to activate TEAS and initiate the SOBER intervention, we see potential for the system to become more proactive in its integration into a person's life. This could be accomplished by integrating mobile data like geographic location, time, and movement, and by allowing users to identify trigger spaces or times which might result in higher chance of relapse. In addition, the system could be further personalized to an individual through readings of their biometrics such as Electrodermal Activity (EDA), Heart Rate (HR), and body temperature, which have all been associated with drug related cue-reactivity [6, 8, 19, 23, 28]. In order to do this, however, we believe that we need to gain a deeper understanding of the correlation between physiological signals and drug related cue-reactivity, and the range of responses one can have to environmental triggers.

Both acupoint stimulation and cue-reactivity are highly personalized. Everything from skin conductance, sensitivity, size, and personal triggers potentially may influence the comfort of the wearable device and strength of acupoint stimulation. For greater personalization in the strength of the acupoint stimulation, the next iteration of the Acu.ation wearable device would include the ability to adjust the pulse amplitude between 10-15 mA. Increased customization can make the device more attentive to individual needs but can also raise costs and decrease accessibility, which should be considered as the product is developed further.

We also recognize that with any self-administered intervention comes the potential for individuals to develop dependence. They may begin to feel that they need TEAS in all circumstances of cue-reactivity in order to avoid relapse. We hope to mitigate this risk by always pairing TEAS with a cognitive behavioral intervention that can serve as a long-lasting support in moments of cue-reactivity. Too frequent self-administration could also lead to the development of tolerance for the stimulation, causing the individual to seek higher and higher intensity. We plan to address this by limiting the frequency and intensity of TEAS that the wearable can deliver. However, we believe that these concerns should also be accounted for by ensuring that an individual is engaged in other relapse prevention strategies alongside this one, and ensuring that there are regulations around how many wearable devices a person can purchase.

Lastly, we feel that it is important to test the Acu.ation prototype with potential users and iterate on the design accordingly, to ensure that the intervention is user friendly and effective. Therefore, the next step in this research is to conduct a lab-based pilot study to test the intervention within the context of cue-reactivity. For this study, subjects will be active smokers who wish to quit, or individuals who have recently quit. Participants would be split into four groups, all of whom will be exposed to a smoking-related cue, and will then wear the device and go through the SOBER technique. The first group will receive no electrical stimulation (0 mA). The second group will receive 5mA of electrical stimulation. The third group will receive 10-15mA of electrical stimulation on the target acupoints. The fourth group will receive 10-15mA of electrical stimulation on non-acupoint locations (locate the acupoints higher or lower than those specified from previous studies [7]). This will allow us

to determine the efficacy of the electrical stimulation and specific electrode placements, and also understand the comfort and appeal of the device for users. After this pilot, we will make any required adjustments to the intervention before continuing with an in-the-wild study with participants actively looking to quit smoking, to explore how well the device integrates into people's lives.

5 CONCLUSION

The use of tobacco products remains pervasive throughout the United States, with smoking-related environmental cues playing a significant role in the urge to smoke and rates of relapse [21, 26, 27, 29]. Cognitive behavioral interventions like MBRP have been shown to reduce rates of relapse but not reduce cue-reactivity [15, 18, 25, 34]. This may prevent people from engaging with cognitive behavioral strategies when they experience particularly strong cue-reactivity, like in the early stages of quitting. Novel studies have found that transcutaneous electric acupoint stimulation (TEAS) can provide immediate relief in the moment of cue reactivity [3, 10, 16]. For this reason, we have designed a wearable device that pairs TEAS with MBRP prompts, to reduce cue-reactivity and increase engagement with helpful relapse-prevention strategies.

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